

Quantitative Finance Q&A

ABSTRACT: This is a list of questions which were asked in the interviews for quant jobs.

1 ABN AMRO

First round, 1 hour, 26th April 2022

1. I have a coin and toss it 1000 times. How can I test whether this coin is fair or not?
2. Imagine we think that the outcome depends on some external parameter, *e.g.*, the wind speed. How can we determine whether this assumption is correct?
3. We can address the above problem by using logistic regression. Consider the sigmoid function

$$\frac{1}{1 + e^{\beta_0 + \beta_1 w}} . \quad (1.1)$$

Should we keep the intercept β_0 to be non-zero?

4. How can you do a simulation to find the value of π ? What about finding π^2 from this setup? What is the error?

Answer to the third part: $\sim 1/\sqrt{N}$, where N is the number of simulations. This is the fluctuation/dissipation universality ‘relative fluctuations reduce as the square root of the number of identical processes’. See around equation (1.58) of Krauth’s book *Algorithms & Computation*.

Second round, 1 hour, 5th May 2022

1. Consider an asset that undergoes geometric Brownian motion

$$dA = \sigma A dW . \quad (1.2)$$

The asset starts at time $t = 0$ for which $A(0) = 1$. What is the probability that the asset price at maturity $A(T)$ is below a certain value ℓ ?

2. What is the probability that the asset price $A(t)$ hits/reaches the value ℓ for the first time at time t ?

Answer: This question has to do with first passage/hitting time. See page 132 of Zhou or Kakushadze’s *Phynance* page 22 or Sec 3.6-7 of Shreve’s vol 2.

3. You roll a dice until the game stops. If you roll 1, 2 or 3, you get the number of dots on the upturned face and the game stops. If you roll 4, 5 or 6 you get the number of dots on the upturned face and you get to roll again. What is the expected payoff?

Same problem as ‘Dice game’ in Zhou’s page 94.

4. What is the total number of squares which you can construct from a chessboard out of the white and black squares?

2 Squarepoint Capital

First round, 1 hour, 1st August 2022

1. **Binge watching** A friend of Alex has gifted a movie collection, and Alex is excited to watch them all as quickly as possible. The duration of the movies given in array `durations[n]`, where `n` is the number of movies, and each movie duration lies between 1.01 and 3.00 units of time (up to two decimal places).

Every day, Alex wants to spend no more than 3.00 units of time watching the movies but also wants to complete the movies in the least number of days possible. Alex does not leave a movie in between. That is, if Alex has picked up a movie, Alex watches the complete movie on the same day. Find the minimum number of days needed to watch all the movies.

Example:

`n = 5`

`durations = [1.90, 1.04, 1.25, 2.5, 1.75]`

(Answer: 3 days)

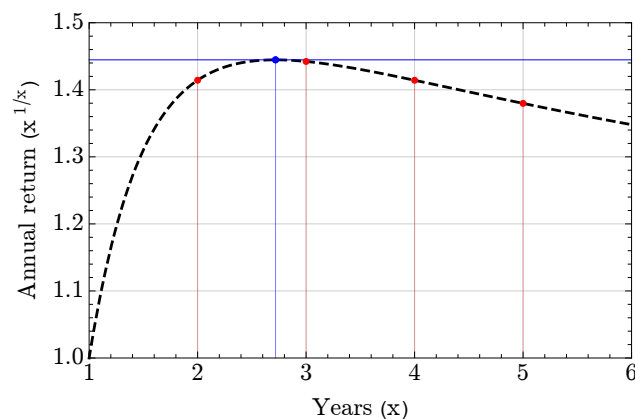
2. You can given an array and you do not know the length of the array. How can you select a random element from the array? You are allowed to make only a single pass through the array.

Second round, 1 hour, 5th August 2022

1. You have three financial instruments. You invest \$1 in all three of them start with. The first one gives you \$2 after 2 years, the second \$3 after 3 years and the third \$4 after 4 years. Which instrument would you choose?

Suppose you want to build an instrument that would return x after x years. How would you do it?

Answer: The point is to realize that the annual return is $x^{1/x}$ and it is maximized by $x = e$. Here is a plot.



2. A and B play a game with a 6-faced die. Whoever rolls 6 first wins. A rolls first, followed by B. What is the probability of A winning the game?
3. Consider you have the historical data of the AAPL stock for the past 10 years. It is seen that if you buy a stock on Wednesday and sell it on Thursday you usually get a profit. Would you invest in this strategy?

Knowing the information for the above stock prices. What else can you do make more money?

3 Heard through the grapevine

1. **Question:** You are in a dark room with two doors. One door leads to another dark room with similar doors, and another door leads to freedom. What is the expected number of rooms you will visit before you are free?

Answer: The question is effectively the same as: what is the expected number of tosses until you see a head? Answer: 2. $P(\text{door to another room})=1/2$, $P(\text{door to freedom})=1/2$. Probability of finding the door to freedom after $(k-1)$ failures is $(1/2)^{k-1}(1/2)^1$. This is the geometric distribution for which $\mathbb{E}(X) = 1/p = 2$.

Another approach: Let e be expected number of rooms we visited. Note that in our first trial we can have the door to freedom with probability $1/2$ (success) or the door to the dark with probability $1/2$ and we are back to the start again. Therefore

$$e = \frac{1}{2} \cdot 1 + \frac{1}{2} \cdot (e + 1) \implies e = 2 . \quad (3.1)$$

The latter approach is useful for problems of the type: what is the expected number of tosses to see three heads in a row?

2. **Question:** Suppose N people all toss a hat into a box and then proceed to randomly pick out a hat. What is the expected number of people to get their own hat back?

Answer: There are N hats in the box and the probability that 1 person finds his own hat is $p = 1/N$. The probability of not finding one's own hat is $q = 1 - p$. Therefore the probability that m people find their own hat is given by the binomial distribution

$$P(m) = \binom{N}{m} p^m q^{N-m} . \quad (3.2)$$

The mean of the binomial distribution is Np . As $p = 1/N$, we have

$$\mathbb{E}[m] = Np = N \times 1/N = 1 . \quad (3.3)$$

This can also be seen through a direct calculation

$$\mathbb{E}[m] = \sum_{m=0}^N m \binom{N}{m} p^m q^{N-m} = p \partial_p (p+q)^N = Np(p+q)^N = Np . \quad (3.4)$$

3. **Question:** You roll a die N times. What is the probability that the maximum of the rolls is m ?

Answer: If a die is rolled N times there are 6^N outcomes. The number of outcomes which consists of numbers in between 1 and m is m^N . The number of outcomes which consists of numbers in between 1 and $m-1$ is $(m-1)^N$. Therefore, the number of outcomes which contains m at least once amongst the N rolls is $m^N - (m-1)^N$. Finally, the probability that the maximum upturned face is m is given by $\frac{m^N - (m-1)^N}{6^N}$.

4. **Question:** Choose N numbers from the uniform distribution $[0,1]$ until we reach a total of ≥ 1 . What is the average value of N ?

Answer: The sum of M uniform random variables (each in the range $[0,1]$) is another uniform distribution but in the range $[0, M]$. The probability density function is $\frac{dx}{M-0}$. Let's call the sum of M random variables as S_M . The probability of M numbers, *and not less*, summing to 1 is given by the product of the following cumulative distributions

$$\begin{aligned} P(S_M > 1 | 1 < S_{M-1} < S_{M-2} < \dots < S_2 < 0) \\ &= P(S_M > 1)P(1 < S_{M-1} < S_{M-2} < \dots < S_2 < 0) \\ &= \left(\int_1^M \frac{dS_M}{M} \right) \left(\int_0^1 \frac{dS_{M-1}}{M-1} \int_0^{1-S_{M-1}} \frac{dS_{M-2}}{M-2} \int_0^{1-S_{M-1}-S_{M-2}} \frac{dS_{M-3}}{M-3} \dots \int_0^{1-S_{M-1}-S_{M-2}-\dots-S_3} \frac{dS_2}{2} \right) \\ &= \frac{M-1}{M} \frac{1}{M-1} \frac{1}{M-2} \dots \frac{1}{2} = \frac{1}{M} \frac{1}{(M-2)!} \end{aligned}$$

Note that we need at least two draws to get a sum > 1 , no matter what. Therefore the minimum expected number of draws to get a sum > 1 is

$$\langle M \rangle = \sum_{M=2}^{\infty} M \times P(S_M > 1 | S_{m < M} < 1) = \sum_{M=2}^{\infty} \frac{1}{(M-2)!} = \sum_{p=0}^{\infty} \frac{1}{p!} = e$$

4 Other random questions

1. What is the number of ways of distributing M identical balls in N identical boxes?

Answer: This is given by the number of partitions of M into at most N integers. The generating function is

$$\prod_{n=1}^M \frac{1}{(1-q^n)} . \quad (4.1)$$

We need to read off the coefficient of q^M of the GF above.

2. What is the number of ways of distributing M identical balls in N distinguishable/distinct boxes?

Answer: If some of the boxes are allowed to remain empty then

$$\binom{N+M-1}{N-1} \quad (4.2)$$

If none of the boxes are allowed to be empty then

$$\binom{M-1}{N-1} \quad (4.3)$$

3. What is the number of ways of distributing M types of balls in N distinguishable/distinct boxes/compartments?

Answer: For $M > N$

$$M^N \quad (4.4)$$

4. What is the number of ways of distributing M distinguishable/distinct balls in N distinguishable/distinct boxes/compartments so that the i 'th compartment contains n_i balls?

Answer:

$$\frac{M!}{n_1!n_2!\cdots n_N!} \quad (4.5)$$

5. How many ways are there to divide M people into pairs assuming there are an even number of people?

Answer:

$$(M-1)!! = (M-1) \times (M-3) \times \cdots \times 1 \quad (4.6)$$

References